MATH IS EVERYWHERE

A Fun Guide to Numbers in Real Life (Ages 8-13)

15 Chapters * Real-World Math * Puzzles * Tricks * Activities

By Daniel Tesfamariam

MATH IS EVERYWHERE: A Fun Guide to Numbers in Real Life

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Published for Amazon KDP

Paperback Edition - 8.5 x 11 inches

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CHAPTER 1

Math in Cooking

Fractions are everywhere in the kitchen! When you follow a recipe, you use fractions: 1/2 cup of sugar, 3/4 teaspoon of salt, 1/3 of the dough. When you double or halve a recipe, you multiply or divide fractions. Understanding fractions through cooking makes them real and delicious!

WOW FACT:

If a recipe serves 4 and you need to feed 6, what do you multiply by? If you need 3/4 cup of flour but you’re doubling the recipe, how much do you need? These are real-world fractions you’ll use for life!

CHAPTER 1: PRACTICE

Problem 1:

You have 3/4 cup flour and need to double it. How much? (3/4 x 2 = ?)

Answer: ____________________

Problem 2:

A recipe serves 8. You want to serve 4. Cut all ingredients by what fraction?

Answer: ____________________

Problem 3:

If you eat 2/8 of a pizza, what fraction is that in simplest form?

Answer: ____________________

Problem 4:

A cake needs 2 1/2 cups sugar. You're making 3 cakes. Total sugar needed?

Answer: ____________________

Problem 5:

You have 1/3 of a bag of chocolate chips. You need to split it 4 ways. Each person gets?

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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CHAPTER 2

Math in Sports

Sports are FULL of math! Batting averages in baseball are decimals. Basketball stats use percentages. Football uses geometry (angles of throws). Soccer uses probability. Even fantasy sports require statistical thinking! Athletes and coaches use math constantly to improve performance.

WOW FACT:

If a basketball player makes 7 out of 10 free throws, their shooting percentage is 70%. If they shoot 100 free throws in a season, you'd predict they'd make about 70!

CHAPTER 2: PRACTICE

Problem 1:

A baseball player gets 3 hits in 10 at-bats. What’s their batting average?

Answer: ____________________

Problem 2:

A team won 15 of 20 games. What’s their win percentage?

Answer: ____________________

Problem 3:

A runner’s times for 5 races: 12.1, 11.8, 12.3, 11.9, 12.0. What’s the average?

Answer: ____________________

Problem 4:

A quarterback throws 25 passes, completes 15. Completion rate?

Answer: ____________________

Problem 5:

A soccer goalie blocks 8 of 12 shots. Save percentage?

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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CHAPTER 3

Math in Shopping

Every time you shop, you use math! Percentages help with sales (30% off!), unit prices help you compare deals, tax adds to your total, and budgeting requires addition and subtraction. Being good at shopping math saves you REAL money throughout your entire life.

WOW FACT:

A \$50 video game is 20% off. That's \$50 x 0.20 = \$10 savings. New price: \$40. But with 8% tax: \$40 x 1.08 = \$43.20. Mental math like this makes you a smart consumer!

CHAPTER 3: PRACTICE

Problem 1:

A shirt costs $25 and is 40% off. What's the sale price?

Answer: ____________________

Problem 2:

Which is cheaper per ounce: 12 oz for $3.60 or 16 oz for $4.48?

Answer: ____________________

Problem 3:

You have $100. After buying items for $23, $17, and $45, what's left?

Answer: ____________________

Problem 4:

A $80 item with 25% off, then 10% tax. Final price?

Answer: ____________________

Problem 5:

If you save $5/week, how many weeks to buy a $75 game?

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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CHAPTER 4

Math in Building

Every building, bridge, and structure uses geometry! Right angles keep walls straight. Triangles provide strength. Circles create domes and arches. Area tells you how much material you need. Volume tells you how much space something holds. Architects and engineers are geometry experts!

WOW FACT:

To build a treehouse with a floor that's 8 feet by 6 feet, you need to know the area: 8 x 6 = 48 square feet of wood. Want to add diagonal bracing? Use the Pythagorean theorem!

CHAPTER 4: PRACTICE

Problem 1:

Find the area of a rectangular room that's 12 ft by 10 ft.

Answer: ____________________

Problem 2:

A circular garden has radius 5 ft. What's its area? (pi x r squared)

Answer: ____________________

Problem 3:

A triangle has base 6 cm and height 8 cm. Area? (1/2 x base x height)

Answer: ____________________

Problem 4:

A box is 3ft x 2ft x 4ft. What's the volume?

Answer: ____________________

Problem 5:

A right triangle has legs 3 and 4. What's the hypotenuse? (a2 + b2 = c2)

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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CHAPTER 5

Math in Nature

Nature is secretly mathematical! Sunflower seeds grow in Fibonacci spirals. Snowflakes have 6-fold symmetry. Honeybees build hexagonal cells (most efficient shape). Tree branches follow mathematical patterns. The golden ratio appears in shells, hurricanes, and galaxies!

WOW FACT:

The Fibonacci sequence: 1, 1, 2, 3, 5, 8, 13, 21, 34... Each number is the sum of the two before it. This pattern appears in flower petals, pine cones, and pineapple scales!

CHAPTER 5: PRACTICE

Problem 1:

Continue the Fibonacci sequence: 1, 1, 2, 3, 5, 8, 13, 21, ?, ?, ?

Answer: ____________________

Problem 2:

Count petals on 5 flowers. Are any Fibonacci numbers? (3, 5, 8, 13, 21)

Answer: ____________________

Problem 3:

How many sides does a snowflake have? What about a honeycomb cell?

Answer: ____________________

Problem 4:

Find the pattern: 2, 4, 8, 16, ___, ___, ___

Answer: ____________________

Problem 5:

Find a spiral in nature. How does it relate to math?

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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CHAPTER 6

Math in Music

Music IS math! Rhythm is fractions (whole notes, half notes, quarter notes). Frequency determines pitch (doubling frequency = one octave higher). Time signatures are fractions. Even harmonious chords are mathematical ratios! Musicians use math whether they know it or not.

WOW FACT:

In 4/4 time, each measure has 4 beats. A whole note = 4 beats. A half note = 2. A quarter note = 1. A measure could be: 1 half note + 2 quarter notes = 2 + 1 + 1 = 4 beats!

CHAPTER 6: PRACTICE

Problem 1:

How many quarter notes fit in one measure of 4/4 time?

Answer: ____________________

Problem 2:

If a song is 3 minutes at 120 beats per minute, total beats?

Answer: ____________________

Problem 3:

A half note + a quarter note + ___ = one full measure (4/4 time)?

Answer: ____________________

Problem 4:

If you practice 30 min/day for 365 days, how many hours total?

Answer: ____________________

Problem 5:

Concert A is 440 Hz. One octave up is double. What frequency?

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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CHAPTER 7

Math in Art

Artists use math for symmetry, proportion, perspective, and patterns! The golden ratio (1.618) creates aesthetically pleasing compositions. Tessellations (repeating patterns with no gaps) are pure geometry. Even color mixing uses mathematical proportions. Art and math are creative partners!

WOW FACT:

Symmetry means both sides match. A butterfly has bilateral symmetry (one line of symmetry). A snowflake has 6 lines of symmetry. A circle has infinite lines of symmetry!

CHAPTER 7: PRACTICE

Problem 1:

Draw a shape with exactly 1 line of symmetry, 2 lines, and 4 lines.

Answer: ____________________

Problem 2:

Create a tessellation using only triangles (no gaps, no overlaps).

Answer: ____________________

Problem 3:

What fraction of a color wheel is red? Blue? Yellow?

Answer: ____________________

Problem 4:

Scale up a drawing: if original is 2x3 inches, what’s 3x bigger?

Answer: ____________________

Problem 5:

Draw a simple picture using only geometric shapes.

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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CHAPTER 8

Math in Games

Games use probability everywhere! Rolling dice, drawing cards, spinning spinners - all probability. Understanding odds helps you make better decisions in games (and life!). Expected value tells you which choices give the best average outcome over time.

WOW FACT:

When you roll one die, the probability of getting a 6 is 1/6 (about 17%). With two dice, the probability of getting a total of 7 is 6/36 = 1/6. Seven is the most common roll!

CHAPTER 8: PRACTICE

Problem 1:

What’s the probability of flipping heads on a coin? (express as fraction)

Answer: ____________________

Problem 2:

Rolling a die: what’s the chance of getting an even number?

Answer: ____________________

Problem 3:

Drawing from a deck: chance of getting a heart? A face card?

Answer: ____________________

Problem 4:

If you flip 3 coins, how many possible outcomes? List them.

Answer: ____________________

Problem 5:

Rolling 2 dice: which total (2-12) is most likely? Why?

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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CHAPTER 9

Math in Travel

Road trips use math constantly! Distance = speed x time. If you’re going 60 mph, you’ll cover 180 miles in 3 hours. Time zones require addition/subtraction. Currency exchange uses multiplication. Map scales convert inches to miles. Travel is one big math adventure!

WOW FACT:

If your car goes 60 mph and your destination is 300 miles away, travel time = 300/60 = 5 hours. If you leave at 9 AM, you arrive at 2 PM!

CHAPTER 9: PRACTICE

Problem 1:

If driving 55 mph for 4 hours, how far do you travel?

Answer: ____________________

Problem 2:

A flight leaves at 3PM Eastern, takes 5 hours. Pacific time arrival? (3 hours behind)

Answer: ____________________

Problem 3:

A map scale is 1 inch = 50 miles. Two cities are 3.5 inches apart. Real distance?

Answer: ____________________

Problem 4:

$1 USD = 0.85 Euros. How many euros for $100?

Answer: ____________________

Problem 5:

A car uses 1 gallon per 30 miles. For 450 miles, how many gallons needed?

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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CHAPTER 10

Math in Money

Compound interest is the most important math concept for building wealth! If you invest $100 at 10% annual interest: Year 1 = $110, Year 2 = $121, Year 3 = $133.10. The interest earns interest! This is why starting to save young is SO powerful - time is your greatest ally.

WOW FACT:

The Rule of 72: divide 72 by your interest rate to find how many years it takes to DOUBLE your money. At 8% interest, 72/8 = 9 years to double!

CHAPTER 10: PRACTICE

Problem 1:

If you invest $100 at 10% interest, how much after 1 year? After 2 years?

Answer: ____________________

Problem 2:

Using Rule of 72: at 6% interest, how many years to double your money?

Answer: ____________________

Problem 3:

You save $10/week for 1 year. Total saved? If it earns 5% interest, year-end total?

Answer: ____________________

Problem 4:

A bank offers 3% vs 5% interest on $1000. Difference after 1 year?

Answer: ____________________

Problem 5:

If $500 doubles every 9 years, how much in 27 years?

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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CHAPTER 11

Math in Space

Space involves ENORMOUS numbers! The Sun is 93 million miles away. Light travels 186,000 miles per SECOND. The Milky Way has 200 billion stars. These astronomical numbers require scientific notation and help us grasp the incredible scale of the universe.

WOW FACT:

Light travels at 186,000 miles/second. In one year, light travels about 5.88 TRILLION miles. That distance is called a light-year. The nearest star is 4.2 light-years away!

CHAPTER 11: PRACTICE

Problem 1:

Light takes 8 min to reach Earth from the Sun. Distance = 186,000 x 480 sec = ?

Answer: ____________________

Problem 2:

Jupiter is 5 times farther from the Sun than Earth (93M mi). Jupiter's distance?

Answer: ____________________

Problem 3:

If a rocket goes 25,000 mph, how many hours to reach the Moon (238,900 mi)?

Answer: ____________________

Problem 4:

Write one million in scientific notation (10 to the what power?)

Answer: ____________________

Problem 5:

The Milky Way has 200 billion stars. Write in scientific notation.

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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CHAPTER 12

Math Puzzles & Riddles

Math puzzles make your brain stronger! They teach creative problem-solving, logical thinking, and persistence. Some puzzles that stumped adults for years were solved by kids who thought differently. Try these 20 puzzles - some are tricky!

WOW FACT:

A classic: A bat and ball cost $1.10 total. The bat costs $1 more than the ball. How much does the ball cost? (Hint: it’s NOT $0.10!)

CHAPTER 12: PRACTICE

Problem 1:

I am thinking of a number. I double it, add 6, and get 20. What’s my number?

Answer: ____________________

Problem 2:

If 3 cats catch 3 mice in 3 minutes, how many cats catch 100 mice in 100 min?

Answer: ____________________

Problem 3:

A farmer has 17 sheep. All but 9 escape. How many are left?

Answer: ____________________

Problem 4:

Fill in: 1, 4, 9, 16, 25, ___, ___, ___ (what’s the pattern?)

Answer: ____________________

Problem 5:

You have 8 balls. One is heavier. Using a balance scale twice, find it. How?

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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CHAPTER 13

Math Tricks to Impress Friends

These 10 tricks will make you look like a math genius! They all use mathematical properties in clever ways that seem like magic but are actually just smart math.

WOW FACT:

The 9 Times Table Trick: Hold up 10 fingers. To multiply 9 x 4, put down finger #4. You have 3 fingers on the left and 6 on the right. Answer: 36!

CHAPTER 13: PRACTICE

Problem 1:

Multiply any 2-digit number by 11: e.g., 36 x 11 = put 3+6=9 in middle = 396!

Answer: ____________________

Problem 2:

To square any number ending in 5: 35 squared = 3x4=12, then add 25 = 1225

Answer: ____________________

Problem 3:

Divisibility by 3: if digits add up to a multiple of 3, the number is divisible by 3

Answer: ____________________

Problem 4:

Mental math: to add 99, add 100 then subtract 1. E.g., 456+99 = 556-1 = 555

Answer: ____________________

Problem 5:

Percent trick: x% of y = y% of x. So 8% of 50 = 50% of 8 = 4!

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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CHAPTER 14

Famous Mathematicians

Math was built by brilliant people from all over the world! From ancient civilizations to modern geniuses, mathematicians discovered the rules that govern our universe. Their stories show that math talent comes in all shapes, sizes, genders, and backgrounds.

WOW FACT:

Katherine Johnson (1918-2020) calculated trajectories for NASA space missions by HAND. Her math sent astronauts to the Moon! The movie ’Hidden Figures’ tells her incredible story.

CHAPTER 14: PRACTICE

Problem 1:

Research one mathematician and write 5 facts about them

Answer: ____________________

Problem 2:

Pythagoras, Euclid, Fibonacci, Newton, Euler, Gauss, Ramanujan, Katherine Johnson

Answer: ____________________

Problem 3:

Ada Lovelace wrote the first computer program in 1843!

Answer: ____________________

Problem 4:

Maryam Mirzakhani was the first woman to win math’s top prize (Fields Medal)

Answer: ____________________

Problem 5:

Who is your favorite? What did they discover?

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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CHAPTER 15

Your Math Adventure Journal

This final chapter is YOUR space to practice math in the real world! Look for math everywhere you go this month. Record what you find, solve problems you encounter, and create your own math challenges. The more you see math in daily life, the more natural and fun it becomes!

WOW FACT:

Challenge yourself to find math in at least 3 situations every day for 30 days!

CHAPTER 15: PRACTICE

Problem 1:

Day 1-7: Track math you encounter daily (shopping, cooking, games, etc.)

Answer: ____________________

Problem 2:

Day 8-14: Create 5 word problems based on your real life

Answer: ____________________

Problem 3:

Day 15-21: Teach someone a math trick. Did they learn it?

Answer: ____________________

Problem 4:

Day 22-28: Find math in nature, art, and music around you

Answer: ____________________

Problem 5:

Day 29-30: Write about how math is connected to your future dreams

Answer: ____________________

CHALLENGE PROBLEM:

Create your own real-world math problem based on this chapter:

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MATH CHAMPION CERTIFICATE

Awarded to:

____

For completing this entire book!

Date: __________________

Author: Daniel Tesfamariam

MY MATH JOURNAL - Entry 1

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY MATH JOURNAL - Entry 2

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY MATH JOURNAL - Entry 3

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY MATH JOURNAL - Entry 4

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY MATH JOURNAL - Entry 5

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY MATH JOURNAL - Entry 6

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY MATH JOURNAL - Entry 7

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY MATH JOURNAL - Entry 8

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY MATH JOURNAL - Entry 9

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY MATH JOURNAL - Entry 10

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY MATH JOURNAL - Entry 11

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY MATH JOURNAL - Entry 12

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY MATH JOURNAL - Entry 13

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY MATH JOURNAL - Entry 14

Date: _______________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MATH ACTIVITY #1

CHALLENGE:

Think of 3 new ideas related to this topic and explain each one.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

MATH ACTIVITY #2

CHALLENGE:

Create a mind map connecting everything you learned.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

MATH ACTIVITY #3

CHALLENGE:

Write a letter explaining this topic to a younger child.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

MATH ACTIVITY #4

CHALLENGE:

Design a poster that teaches the most important concept.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

MATH ACTIVITY #5

CHALLENGE:

List 10 questions you still have about this topic.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

MATH ACTIVITY #6

CHALLENGE:

Draw a comic strip showing what you learned in action.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

MATH ACTIVITY #7

CHALLENGE:

Write 5 ways you can apply this knowledge in real life.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

MATH ACTIVITY #8

CHALLENGE:

Create a quiz for a friend (with answer key!).

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

MATH ACTIVITY #9

CHALLENGE:

Compare and contrast two concepts from this book.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

MATH ACTIVITY #10

CHALLENGE:

Write a short story using ideas from this chapter.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

CREATIVE SPACE: MATH

Draw what you learned today

Use this space for drawing, writing, or brainstorming!

CREATIVE SPACE: MATH

Illustrate your favorite concept

Use this space for drawing, writing, or brainstorming!

CREATIVE SPACE: MATH

Create a visual summary of this chapter

Use this space for drawing, writing, or brainstorming!

CREATIVE SPACE: MATH

Draw yourself using this knowledge

Use this space for drawing, writing, or brainstorming!

CREATIVE SPACE: MATH

Design a symbol that represents this topic

Use this space for drawing, writing, or brainstorming!

CREATIVE SPACE: MATH

Sketch your goals

Use this space for drawing, writing, or brainstorming!

NOTES & IDEAS

NOTES & IDEAS

NOTES & IDEAS

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